Jan 2020

Q2 Statistics & Probability.

a) See Notes for all definitions

b) A – strong negative correlation
B – no correlation
C – strong positive correlation

c)

	f_i	M_i	f_iM_i
2-4	3	3	9
4-6	18	5	90
6-8	9	7	63
8-10	7	9	63
	37		225

M = midpoint.

i) Mean: $\bar{x} = \frac{\Sigma f_i M_i}{\Sigma f_i} = \frac{225}{37} = \boxed{6.08}$

ii) Standard deviation.

	f_i	m_i	f_iM_i	M_i^2	$f_iM_i^2$
2-4	3	3	9	9	$3\times9 = 27$
4-6	18	5	90	25	$18\times25 = 450$
6-8	9	7	63	49	$9\times49 = 441$
8-10	7	9	63	81	$7\times81 = 567$
	37		225		1485

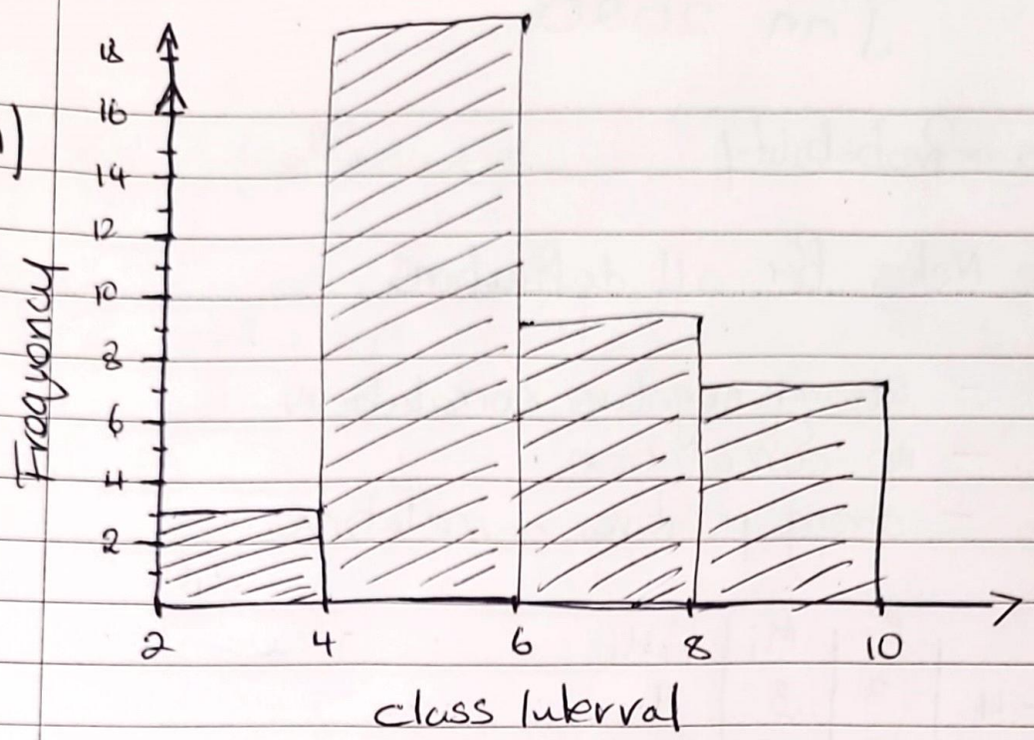
$M = \Sigma f_i$
m = Midpoint

$$S^2 = \frac{\Sigma f_i M_i^2 - M(\bar{x})^2}{M-1} = \frac{1485 - 37(6.08)^2}{37-1}$$

$$S^2 = 3.26$$

$$S = \sqrt{3.26} = \boxed{1.81}$$

iii)



iv) Comment on symmetry / skewness ...

d) Lotto – use combinations (order does not matter)

$n = 32$
$r = 4$

$$^nC_r = \frac{n!}{(n-r)!r!}$$

$$= \frac{32!}{(32-4)!(4!)}$$

$$= \boxed{35,960}$$

e) Binomial use $P(r \text{ successes in } n \text{ trials}) = {}^nC_r\, p^r q^{n-r}$

$n = 40$
$r = 3$
$P = 5\% = 5/100 = 0.05$
$q = 1 - P = 1 - 0.05 = 0.95$

} sub into the formula.

$$P(3 \text{ successes in } 40 \text{ trials}) = \frac{40!}{(40-3)!\,3!} \times 0.05^{3} \times 0.95^{40-3}$$

$$= \boxed{0.18511} \text{ to 5 decimal places.}$$

Another way the question could be phrased —

In a factory, an average $\frac{1}{20}$ out of 20 items produced are defective. In a random sample of 40 items, using the binomial distribution, what is the probability that 3 chosen at random are defective?

$p = 1/20 = 0.05$

$q = 1 - p = 1 - 0.05 = 0.95$

$n = 40$

$r = 3$